

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 318 — Past Questions & Materials
by Renance

Course: EMT 318
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — EMT 318 Methods in Environmental Chemistry II

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

2024/2025 ACADEMIC SESSION FIRST SEMESTER

EMT 318: METHODS IN ENVIRONMENTAL CHEMISTRY II (2 Units)

Instruction: Answer 4 questions, at least one from each section 2 hours

SECTION A

1. (a). Define Colorimeter
- (b). Outline the basic principles of photometry
2. (a). Define the term spectrophotometric titration.
- (b). Enumerate the instruments used in spectrophotometric titration.

SECTION B

3. (a). Why sampling and analysis are an integral part of data quality? Between sampling and analysis, which one often generates more errors? Why?

(b). A new method is being developed for the analysis of a pesticide in soils. A spiked sample with a known concentration of 17.00 mg/kg was measured five times using the new method (data: 15.3, 17.1, 16.7, 15.5, 17.3 mg/kg) and the established EPA method (data: 15.4, 15.9, 16.7, 16.1, 16.2 mg/kg). The Excel outputs of descriptive statistics and one-way ANOVA are given below ($\alpha = 0.05$). On the basis of these output, (a) are these two methods significantly different at a 95% confidence level? (b) Which method is more accurate, why? (c) Which method is more precise, why?

(1) Excel output of descriptive statistics

EPA method		New Method	
Mean	16.06	Mean	16.38
Standard error:	0.211187121	Standard error:	0.412795349
Median	16.1	Median	16.7
Mode	#N/A	Mode	#N/A
Standard deviation:	0.472228758	Standard deviation:	0.923038461

Sample variance:	0.223	Sample variance:	0.852
Kurtosis	1.01288963	Kurtosis	-
Skewness	0.105406085	Skewness	2.910136878
Range	1.3	Range	0.400799173
Minimum	15.4	Minimum	2
Maximum	16.7	Maximum	15.3
Sum	80.3	Maximum	17.3
Count	5	Sum	81.9
Largest(1)	16.7	Count	5
Smallest (1)	15.4	Largest(1)	17.3
Confidence Level (95.0%)	0.586350662	Smallest (1)	15.3
		Confidence Level (95.0%)	1.146106

(2) Excel output of one-way ANOVA

Summary						
Groups	Count	Sum	Average	Variance		
EPA Method	5	80.3	16.06	0.223		
New Method	5	81.9	16.38	0.852		
ANOVA						
Source of variation	SS	df	MS	F	p-value	f-crit
Between Groups	0.256	1	0.256	0.4762791	0.5096343	5.317644991
Within Groups	4.3	8	0.5375			
Total	4.556	9				

- 4 (a). Describe the difference between systematic grid sampling and systematic random sampling using a diagram.
- (b). In a site assessment for the identification of contaminated source, which one of the following is likely the best and the least favorable: (a) judgmental sampling; (b) simple random sampling; and (c) systematic random sampling? Briefly explain
- (c). Given the following particular situation, identify which sampling design approach should be used and discuss why this is the most appropriate?

- i. If you are performing a screening phase of an investigation of a relatively smallscale problem, and you have budget and/or a limited schedule. Your goal is to assess whether further investigation is warranted that should include a detailed follow-up sampling *budget*
- ii. If you are estimating mean and you have known the existence of spatial or temporal patterns of the contaminant. Your goal is to increase the precision of the estimate with the same number of samples, or achieve the same precision with fewer samples and lower cost.
- iii. If you are estimating a population mean and you have an adequate budget. In the meantime, you are also interesting in knowing the information of a spatial or temporal pattern *Sig*
- iv. If you are estimating a population mean and you have budget constrains. The analytical costs are much higher compared with sampling costs. Your goal is to produce an equally precise or a more precise estimate of the mean with fewer analyses and lower cost *and*
- v. If you are developing an understanding of where contamination is present and you have an adequate budget for the number of samples needed, your goal is to acquire coverage of the area of concern with a given level of confidence that you would have detected a hot spot of a given size. *Lim*

If you are developing an understanding of where contamination is present and you have an adequate budget for the number of samples needed, your goal is to acquire coverage of the time periods of interest

Paper 2 of 2 — EMT 318 METHODS IN ENVIRONMENTAL CHEMISTRY II 2024I2025

Level 300 • First Semester • PastQuestion

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(1) Excel output of descriptive statistics

EPA method New Method

Mean 16.06 Mean 16.38

Standard error: | 0.211187121 | Standard error: | 0.412795349

Median 16.1 Median 16.7

Mode #N/A Mode HN/A

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Sample 0.223

Sample 0.852

Variance:

variance:

Kurtosis 1.01288963 Kurtosis - |

| 2.910136878
 Skewness | 0.105406085 Skewness 0.400799173
 _ Range | 1.3 | Range 2 :
 | Minimum 115.4 Minimum 133
 | Maximum 16.7 Maximum 11723
 3 Sum 80.3 Sum 81.9
 S | Count "5 | Count 5
 " i |
 x | Largest(1) 16.7 Largest(1) igs
) f
 | Smallest (1) | 15.4 Smallest (1) 15.3
 | Confidence | 0.586350662 | Confidence 1.146106 |
 | Level (95.0%) | Level (95.0%) |
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